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TEXtured Manual

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Introduction

Already when I started writing my bachelor thesis [1], I tweaked a lot the original LaTeX template (which itself was slightly updated since then [2]) provided for students at MFF CUNI — Faculty of Mathematics and Physics, Charles University in Prague.

Some design choices originated already during this stage, particularly the significant use of theorem-like and remark-like environments with highly interlinked structure of the text.

I picked up right where I left off when I started writing my master thesis [●], and I have been refining the template ever since. Improved understanding of the coding backbone behind LaTeX and its package ecosystem enabled me to customize it even further to my liking, and add even more "bells and whistles".

Remark 1 (Template Purpose)**.** While primarily targeting theses, TeXtured can be used for other document types as well. ⌟

To make it user-friendly, I have restructured the preamble into several files, each of which is responsible for a specific aspect of the document. This way, the user can (and is encouraged to) easily find the relevant part of the code and modify it.

Numerous comments and explanations are provided throughout the code to further aid the user in understanding the template without always having to consult the documentation of packages (which is recommended for more advanced changes).

Remark 2 (How to Setup)**.** To set up TeXtured template for your document, you can use the `Overleaf` template or clone the repository on GitHub [4]. Then, you can start modifying the files to suit your needs.

Also make sure to check the `README.md` file for more detailed instructions, particularly on various software dependencies. See `jdujava/TeXtured #5` if you encounter any issues. ⌟

Remark 3 (MFF CUNI Template Compatibility)**.** TeXtured can be used out of the box for theses at the Faculty of Mathematics and Physics, Charles University in Prague. Just be sure to include all *front matter* pages and fill out necessary data:

- Title Page with the faculty logo (among other things),
- Declaration,
- Dedication (optional),
- Information Page including the Abstract.

This is done by uncommenting the relevant lines in the main file `thesis.tex`.

Layout of these front matter pages is adapted and modified from the original MFF CUNI template [2]. However, always make sure it is compliant with the faculty guidelines, otherwise please raise an issue on GitHub [4]. ⌟

Quick Summary

In **Chapter N** we exhibit an example of a *front matter* chapter.

In **Chapter 1** we present design *Principles* to which TEXtured adheres.

In **Chapter 2** we describe various implemented features, design choices, and LaTeX packages helping with the task of realizing goals sketched in Chapter 1.

In **Chapter 3** we give tips and tricks on how to fully utilize and even extend capabilities of TEXtured.

In **Chapter A** we show an example of an Appendix chapter.

Note (WIP Disclaimer)**.** Both Chapter 2 → p. 2 and Chapter 3 → p. 10 are as of now Work-in-Progress. There is a lot of stuff yet to be exhibited and explained. All the colored TODO-like environments are to be resolved in the final version of this document.

Notation & Conventions

Example N.1 (Usage of Mathematical Fonts)**.** To make the text more readable and beautiful, we can use different types of mathematical fonts for different types of objects (striving to be at least somewhat consistent):

- ***Bold*** often for tensorial object (abstract index).
- Sans for groups, certain spaces, or some operations/maps.
- Fraktur for algebras (and densities).
- *Calligraphic* (available are only capital letters, and ℓ)
- *Calligraphic* (alternative font containing also lowercase letters)
- $\mathbb{D}$ouble-$\mathbb{S}$truck for fields like $\mathbb{R}$, spaces like $\mathbb{S}^n$ and $\mathbb{CP}^n$.
- `Typewriter` for code functions, or other special objects. ⌟

Remark N.2. This chapter is numbered (or perhaps more precisely "lettered"). This means that is appears in Table of Contents with its letter "**N**", which also prefixes all numbering of environments in this chapter.

On the other hand, Introduction → p.iii and Quick Summary → p.iv are unnumbered (or "unlettered") in this sense. ⌟

Example N.3. You can use ◌ as an argument placeholder. ⌟

1 Design Principles

It is not like I stated the *Principles* at the beginning and then tried to follow them. They emerged more naturally. So the causal structure is more like

$$\begin{array}{c}\text{made some}\\ \text{design choices}\end{array} \quad \text{and} \quad \begin{array}{c}\text{implemented}\\ \text{certain features}\end{array} \quad \rightsquigarrow \quad \begin{array}{c}\text{recognized}\\ \text{at first subconscious}\\ \text{overarching principles}\end{array} .$$

Anyway, here they are.

Definition 1.1 (Design Principles)**.** The main design *Principles* are:

- **Elegance** — Aim for a classy, typographically elegant layout.
- **Structure** — Create a smart, easy-to-reference, and skimmable structure.
- **Clarity** — Eliminate distractions and strive for clear explanations.

Remark 1.2 (Common Goal, Alternative Definition via Antiprinciples)**.** There is also an alternative point of view. The common goal of Design Principles is to minimize the following *Antiprinciples*:

- We should be concise, and that means fewer pages, the better. Long blocks of text without noticeable space between paragraphs are preferred (the reader should go on a walk to have some breathing spacetime).
- Avoid creating distinct anchor points for important concepts, since an attentive reader should be able to extract them from blocks of text.
- Do not waste time referencing earlier discussions and reflecting on them from the current context and point of view, as the reader is anyway making such connections all the time. ⌟

Each of these principles is somehow reflected in the design choices and features included (or omitted) in TEXtured, see Chapter 2 →p.2 for more details.

Remark 1.3 (Disclaimer)**.** The following is at places highly opinionated, and not applicable to all scenarios and use-cases. I tried to describe my reasons for specific design choices, with which you can certainly disagree. I hope that at least it can provoke more people *(especially you!)* to contemplate about document creation, ideally resulting in production of documents with overall better quality. ⌟

2 Features of TEXtured

First two sections are about typography of the document, next are more about its contents and structure.

2.1 Page Layout and Style `preamble/layout/`

Page Dimensions, Printing Layout `preamble/layout/geometry.tex`

`geometry` → CTAN package — set up the page layout (supported single/double-sided printing). Apply `\flushbottom` — try to make text body on all pages have the same height.

Page Headers and Footers `preamble/layout/headers.tex`

`fancyhdr` → CTAN package — page headers and footers — consistent style also for initial page of a chapter (not totally different style with numbering in the bottom center . . .).

Heading Style `preamble/layout/titles.tex`

Pretty chapter heading style — big calligraphic number/letter behind the title.

2.2 Sane Typographical Defaults `preamble/general/`

Now we will concern ourselves with more intricate and detailed typography (more at level of sentences, words, and even letters).

Paragraphs, Captions `preamble/general/typesetting.tex`

No paragraph indentation, proper space between paragraphs — `parskip` → CTAN.

Floats, Captions `preamble/general/floats.tex`

Caption styling — slight hang, `\footnotesize` font, bold sans label (like other environments).

Font and Related Stuff `preamble/general/typesetting.tex`

The default choice are *Latin Modern* fonts — a classic really. Various families and shapes are typically used for different purposes:

- *Serif* family for the main text

- *Slanted* shape for emphasis using `\emph` macro (instead of the default *Italic* shape, which is reserved mainly for math formulas)

  Remark 2.1 (Nested Emphasis)**.** Nested emphasis is displayed in *Italic* shape. It is rather rare to nest an *additional emphasis inside an emphasis.* ⌟

- *(Bold) Sans* family for headings and other structural elements
- *Typewriter* family for computer code and similar stuff

This is Latin Modern Sans $\alpha = 2^2$
This is Latin Modern Sans Oblique $\alpha = 2^2$
This is Latin Modern Sans Bold $\boldsymbol{\alpha = 2^2}$
This is Latin Modern Sans Bold Oblique $\boldsymbol{\alpha = 2^2}$

This is Latin Modern Sans $\alpha = 2^2$
This is Latin Modern Sans Oblique $\alpha = 2^2$
This is Latin Modern Sans Bold $\boldsymbol{\alpha = 2^2}$
This is Latin Modern Sans Bold Oblique $\boldsymbol{\alpha = 2^2}$

> **Note.** Sans math font has problems with showing properly all bold symbols (sub/superscrips don't work automatically).

For consistent quotation use `\enquote` macro provided by `csquotes`→CTAN.

Micro-Typography `preamble/general/typesetting.tex`

Enable micro-typographic extensions with package `microtype`→CTAN, most prominently character protrusion and font expansion.

Following quote from `microtype`→CTAN documentation nicely explains what it is about:

> Micro-typography is the art of enhancing the appearance and readability of a document while exhibiting a minimum degree of visual obtrusion. It is concerned with what happens between or at the margins of characters, words or lines. Whereas the macro-typographical aspects of a document (i.e., its layout) are clearly visible even to the untrained eye, micro-typographical refinements should ideally not even be recognizable. That is, you may think that a document looks beautiful, but you might not be able to tell exactly why: good micro-typographic practice tries to reduce all potential irritations that might disturb a reader.

2.3 Document Structure

Implemented mostly via `tcolorbox`→CTAN package (and `thmtools`→CTAN).

Custom Structure Environments `preamble/environments/*.tex`

- clear structure, enables high level of interlinking

- include title if possible — rough idea after first look, quicker navigation/finding
- non-intrusive indicators for the end of non-boxed environments
- shared numbering, together with tables, figures, equations, ... — linear increase of reference number, easier to find
- easy to skim through, get quickly an idea, and know what to expect
- not only for physics/math texts, can be used to highlight key ideas
- no emphasis for the whole body of definitions ... we have box for that ...

Remark 2.2. There are also helper environments for Todo-like notes. By default, there are Todo, Note, Suggestion, and Question environments, but you can easily create your own. ⌟

> **Note.** No "code listing" setup yet. PRs welcome.

References, Links `preamble/hacks/custom-reference-boxes.tex`

Custom reference/link/citation styles.

- I like distinguished references/links/citations
- always use `\Cref` from `cleveref` → CTAN package
- frames are useful, but not that pretty (they can obscure punctuation ...), and the PDF viewer must support them
- just color is sometimes alright, but less ideal when printing
- sometimes good to have an idea where they point (mainly when printed, no ability to click) — include page number
- there is slight inconvenience due to small flexibility around line breaks (would be nice to have proper workaround, like link breaking or something like that)

Table of Contents, Index `preamble/layout/toc.tex`

Clear Table of Contents.

- subsections included (in a compact style), but not numbered...
- don't frame references in the Contents, just keep it clean and simple
- include everything important

Proper Index (ToC in PDF viewer).

- very handy for navigating longer documents
- include also other important pages other than just initial pages of main chapters: Contents, Introduction, References, ...
- I use `Zathura` as my PDF viewer, where Index is just one `Tab` away ...

2.4 Bibliography/References

`preamble/references/`

Pretty and functional Bibliography/References, via `biblatex` → CTAN package.

Bibliography Style

`preamble/references/*.tex`

Entries in References → p. 16:

- have a clean consistent style based on the `ext-numeric-verb` style from `biblatex-ext` → CTAN package
- support (new) `github` field
- have DOI/arXiv/URL/GitHub links on the new line `preamble/references/doi-eprint-url.tex`
- include "backlinks" `preamble/references/backref.tex`

Remark 2.3 (Bibliography Data)**.** Make sure to gather all the relevant data you need for every reference. If you later decide you want to reduce the amount of presented information, `biblatex` → CTAN can help you with that. For example, it is possible to automatically

- remove `url` field if `doi` field is present,
- ignore unwanted fields (`pages`, `number`, `volume`, `series`, `location`, ...).

Citation Style

`preamble/references/cite.tex`

Include `[` and `]` characters around citation number inside the link (and wrap in `tcolorbox` → CTAN ...), for example `[4]`.

2.5 PDF/A Compliance

`preamble/pdfA-compliance/`

Proper metadata setup (via `hyperref` → CTAN and `\DocumentMetadata`).

Remark 2.4 (Document Data)**.** Various data about the work should be entered in `preamble/data.tex` file. When the relevant entries contain LaTeX commands (for example to obtain specific formatting of the title), it is necessary to provide "plaintext" variations, so that `hyperref` → CTAN can properly set up PDF metadata.

Next we will describe various common violations of PDF/A standard, and how to fix them.

Glyph to Unicode Map

`.../pdfA-compliance/glyphtounicode.tex`

To obtain PDF/A compliant PDF, we need to have Unicode mapping for all glyphs used in the document. It can happen — mainly when using fonts providing extra mathematical symbols — that certain glyphs are not covered by mappings loaded in `preamble/pdfA-compliance/glyphtounicode.tex`.

In the `preamble/pdfA-compliance/glyphtounicode.tex` file you can also find an example veraPDF output for a PDF with a problematic glyph. It also points to a guide located in `preamble/pdfA-compliance/LaTeX-find-glyph-name/` directory, which explains how to find out the glyph name, and how to provide the *glyph to Unicode* mapping with `\pdfglyphtounicode` command.

PDF `/Interpolation` Key

Some PDFs can have enabled the `/Interpolation` key, for example `Inkscape` generated PDFs with blur parts. However, PDF/A requires it to be disabled.

This is automatically fixed by `figures/Inkscape/inkscape-export-to-latex` shell script.

2.6 Code Organization

- code broken into multiple files by its function, see Figure 2.6 → p. 7
- use `aux_dir` to minimize clutter
- avoid large, hard to navigate, preamble files
- try to follow "one sentence per line" rule — improves diffs for versioning systems like `git` — tools like `latexindent` can help

> **Note.** My config for `latexindent` mostly works, but some corner cases can surface. Will share someday.

Remark 2.5 (Custom User Macros)**.** Enter your own macros in the `preamble/user/` directory, which is reserved precisely for this purpose. Then, if you would like to update to a newer version of TEXtured, you will be having easier time — less mixing of your code with the template code will result in fewer conflicts you must resolve manually. ⌟

2.7 Miscellaneous

Math-Related Tweaks — $e^{i\pi}$ `preamble/math/`

Some of the math-related tweaks:

- Use `\boldmath` automatically for `\textbf` text (useful mainly in headings).
- Possible to use sans italic font for math via `\mathsfit`.
- Better extendable arrows with `TikZ` → CTAN.
- *(Optional, disabled by default)* Automatically change usage of `\textcolor` to `\mathcolor` in math mode, so that we get proper math spacing, for example

$$a \times b \text{ (right spacing)} \qquad \text{versus} \qquad a \times b \text{ (wrong spacing)}\,.$$

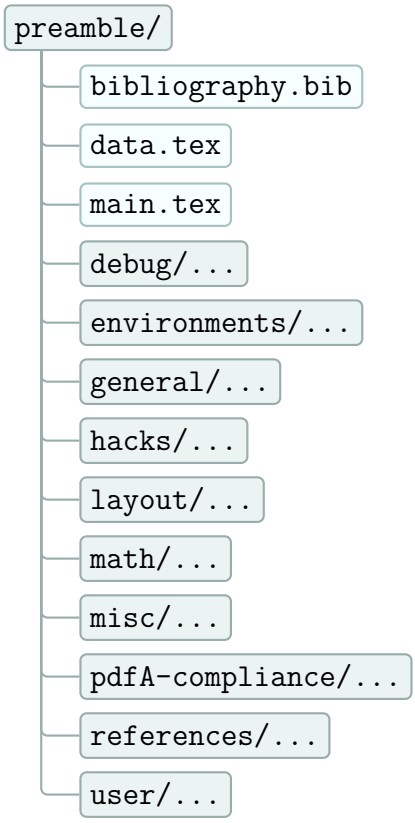

Figure 2.6 / File structure of the `preamble/` folder.

Make sure to fill all (meta)data — like title, author, etc. — in the `preamble/data.tex` file. The bibliography/reference data is stored in the `preamble/bibliography.bib` file.

Apart from the `preamble/pdfA-compliance/glyphtounicode.tex` file, which needs to be sourced before calling `\documentclass`, all other files are loaded in the `preamble/main.tex` file.

When possible, add your own tweaks and macros to the `preamble/user/` directory reserved for this purpose. This way, you can easily update to newer versions of TEXtured (hopefully) without conflicts.

However, it is recommended to explicitly use `\mathcolor` when appropriate, since it leads to easier maintenance of the code (copy-pasting to other projects will work without problems).

Following practice is highly recommended.

Tip 2.7 (Define Your Own Math Macros)**.** Frequently define macros for notation used more than once. Advantages are for example:

- Code is easier to read/write, since it is more "semantic".
- To tweak notation, you only need to change it in one place.
- Easier to find all occurrences of a certain notion. ⌟

GitHub Actions

`.github/workflows/`

Describe implemented GitHub Actions:

- Automatic `latexmk` build of the latest PDF version.
- PDF/A verification via veraPDF.
- Deploy to `gh-pages` branch. One can furthermore enable (in repo settings) GitHub Pages for `gh-pages` branch, which will automatically upload latest PDF to `https://username.github.io/reponame/thesis.pdf`. This enables convenient sharing of your (even continuously evolving) work without needing to commit the PDF (resulting in large repository size) or compiling the PDF on

the receiving side.

Remark 2.8 (Private Repositories)**.** Even for private repositories such link is publicly accessible. This is why GitHub Pages setup is not done automatically for you. If you want to share the work more "privately", there are other solutions, for example GitHub Action which uploads PDF to Google Drive, and sharing via a private link. Also look at Section 2.7 → p.8. ⌟

Censoring

`preamble/debug/censor.tex`

Censoring/redaction using `censor` → CTAN package. Use `\censor` or `\censorbox`. For example, censor the following sentence: [redacted].

`Inkscape` Integration

`preamble/misc/inkscape.tex`

- automatic export after changing the `svg` (need to enable `--shell-escape` for pdfTEX or LuaTEX, done via `.latexmkrc`)
- watermark via `ps` injection

  Remark 2.9 (Watermark String)**.** Edit the watermark text in the shell script `figures/Inkscape/inkscape-export-to-latex` to your liking. ⌟

  > **Todo.** Make it easier to change the watermark text. Or extract it somehow automatically from the PDF document?

- automatic fix of `/Interpolation` key problem

2.8 Non-Features

These features were deemed unnecessary, or even counterproductive, and thus were not implemented/not customized. This does not mean that it is hard or not compatible to use them with TEXtured.

Footnotes

- they break the flow of reading, can be distracting
- either it is important and you want it there — no need to use footnotes, or it is not so important (maybe just a reminder/remark), but then there are in my opinion better ways to handle such situation
  - grayed out/smaller text, sidenotes are better alternative, if the page layout enables them
  - it is not bad to remind reader of something in the main text…

List of Figures, List of …

Don't really see the point, but one is free to include it.

Index, Glossary

- since the text is primarily intended for electronic use, finding usage of certain terms is easy
- text should be ideally structured in such a way, that finding definitions of important terms is straightforward — interlinking/referencing in proper places to indicate where the notion to be used was defined/discussed

3 Tips & Tricks

In this chapter we will see how to utilize and extend capabilities of TEXtured. Also, there will be sprinkled miscellaneous tips on how to improve the quality of your document.

3.1 Structure

- numbered and "lettered" chapters

  > **Todo.** Describe `\chapternotnumbered`, and "lettered" chapters in front matter.

- Use nicely named subsections — easier to navigate (also better ToC and Index)

  > **Todo.** Describe `\texorpdfstring`. If you want, you can even use UTF-8 math characters (superscripts/subscripts, emulate math fonts).

- Utilize remark/definition environments to make the document more structured and easier to read. Important concepts will stick out more and will be remembered better.

  > **Todo.** Describe creating of new environments, `\qedhere`, ...

- Try to motivate every definition/theorem with "normal" text, do not let the document degenerate just into a listing of definitions/theorems/proofs/...

- Use references to other remarks/definitions/sections to make the document more interconnected, which can help the reader to look at a bigger picture, recollect necessary information to proceed further, or to understand the context better.

  > **Todo.** Describe `\Cref`, `\Nref`.

  > **Todo.** Show using `\autocite{TODO}` in the text [●]. Helps to not forget to add the citation later.

3.2 Typography

- use `~` to enter non-breakable space, or also after dot in initials/after titles (otherwise one gets bigger space than is proper), for example `M.Sc.~Name Surname`
- proper usage of hyphens/dashes — learn when to use hyphen - (`-`), when en-dash – (`--`), and when em-dash — (`---`)
- use *emphasis* with `\emph` for the names of new and important concepts
- for quotation marks use `\enquote` from `csquotes` → CTAN package

- sometimes using gray text instead of parentheses may result in a cleaner look, for example instead of "(pseudo-)Riemannian" just gray out "pseudo-" like "pseudo-Riemannian"
- choose capitalization style of titles, and stick with it — I choose "titlecase"

3.3 Mathematics & Physics

Math Typesetting

Learn stuff in `amsmath` → CTAN and `mathtools` → CTAN packages. Then it is possible to write stuff like following inclusion map

$$\begin{aligned} \iota\colon (\mathbb{S}^1, \mathbb{R}_{\geq 0}, \mathbb{S}^{d-1}) &\longrightarrow \mathsf{AdS}/\mathbb{Z} \\ (\ t\ ,\ \ r\ \ ,\ \ \overline{\boldsymbol{x}}\ \ ) &\longmapsto \boldsymbol{x} = \iota(t, r, \overline{\boldsymbol{x}}) \equiv \begin{cases} \boldsymbol{x}^{-1} = \sqrt{\ell^2 + r^2}\cos(t/\ell), \\ \boldsymbol{x}^0 = \sqrt{\ell^2 + r^2}\sin(t/\ell), \\ \boldsymbol{x}^i = r\overline{\boldsymbol{x}}^i \quad \text{for} \quad i = 1, \dots, d \end{cases} . \end{aligned}$$

Remark 3.1 (Math Ending Punctuation)**.** Make sure to use `\eqend` or `\eqcomma` macro (when appropriate) to properly end a math environment with a period or a comma, respectively. They add a small space before the punctuation to make the formula look better. ⌟

Todo. Maybe show diagrams with `TikZ` → CTAN package.

Todo. Describe `\DeclareDocumentCommand`, ...

Numbers and Units

Use `siunitx` → CTAN package for convenient typesetting numbers and units. Examples are shown in Table 3.2 → p.11.

Note. The `siunitx` → CTAN package is very powerful and flexible. It can be even used to nicely align numbers in tables. As of now, this feature is not customized in any way in TEXtured. Suggestions for improvements are welcome.

Command	Output	Usage
`\num{123.45 e-8}`	$123.45 \cdot 10^{-8}$	numbers
`\si{\meter\per\second\squared}`	$\mathrm{m/s^2}$	units
`\SI{123.45}{m/s^2}`	$123.45\,\mathrm{m/s^2}$	numbers with units
`\SIrange{1}{10}{\kilo\meter}`	1–10 km	ranges
`\SIlist{1;3;5}{A}`	1 A, 3 A and 5 A	lists
`\SI{1.23 +- 0.45}{\celsius}`	$(1.23 \pm 0.45)\,^\circ\mathrm{C}$	uncertainties

Table 3.2 / Examples of `siunitx` → CTAN package usage.

3.4 LaTeX Coding

Todo. Describe how to create custom macros with `\NewDocumentCommand`, `\RenewDocumentCommand`, `\NewCommandCopy`, ...

Question. Difference between "macro" and "function" in LaTeX? Which nomenclature is appropriate?

Remark 3.3 (Macro Space Handling)**.** Using macro inside text in the form `\foo` can swallow the following whitespace. When this is not the desired behavior, call the macro like `\foo{}`. In this way an empty argument is passed to the macro, leaving the following whitespace intact. ⌟

Todo. Describe `\makeatletter` and `\makeatother`.

Todo. Describe `\ensuremath`. When math macro is used often outside math mode (alone as `...\(\foo\)...`), defining it wrapped in `\ensuremath` can lead to perhaps easier use (as just `...\foo{}...`).

Todo. Describe `\includeonlysmart`.

Note. Be careful about implicit end of line spaces in function definitions, sometimes necessary to use `%` after last command on the line. TODO: Describe this in more detail.

Todo. Describe WIP mode (particularly with LuaLaTeX).

Note. Some comments in source code refer to files from TeXLive installation on Arch Linux. On other distributions or operating systems the paths might be different.

Summary and Outlook

Summary and Outlook.

Example of Appendix Chapter

Example A.1 (Figure Caption Tweaking)**.** Now we will show off some figures with tweaked position/extent of the captions. Figure A.2 → p. 14 has a side-caption, while Figure A.3 → p. 14 has a caption that spans just the width of the figure. This utilizes the `floatrow` → CTAN package and is inspired by ITT LaTeX template [5].

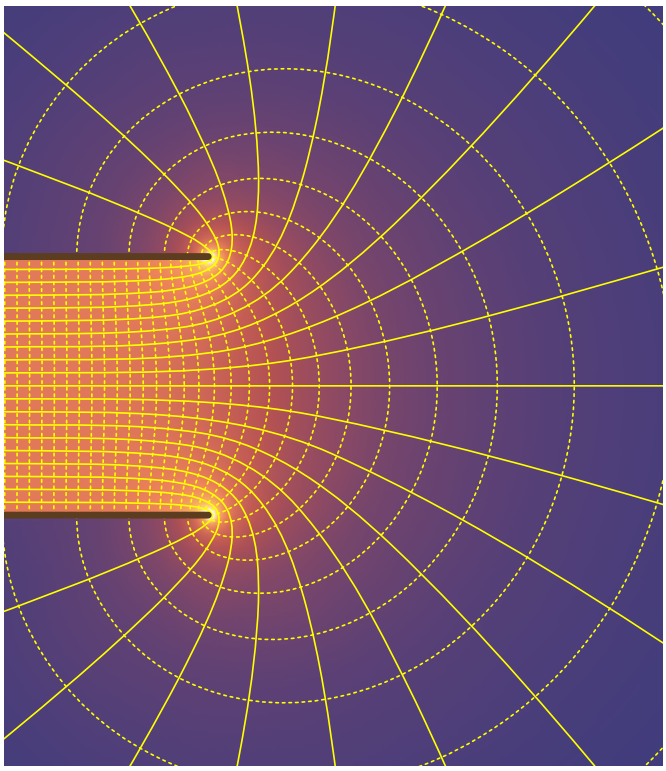

Figure A.2 / Example of a figure with a side-caption.

It displays the two-dimensional electric field near one end of a parallel plate capacitor.

Legend:

- equipotentials
- field lines
- capacitor plate

You can also optionally use a footnote for the figure caption.

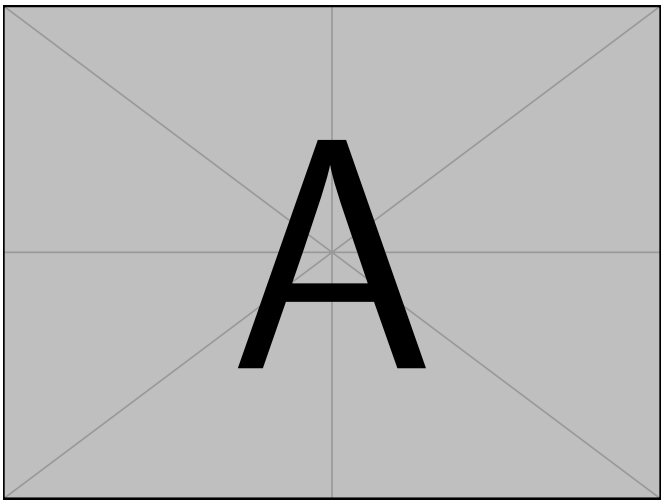

Figure A.3 / Example of a figure with a caption spanning just the width of the figure.

Example A.4 (Multi-Paragraph Figure Caption with Verbatim Text)**.** It is possible to have multi-paragraph captions for figures. One must remember to provide a short description as `\caption[This is a Short Description]{...}`, or else LaTeX will complain.

See Figure A.5 → p. 15 for an example, demonstrating also a workaround for typesetting verbatim text in contexts where "fragile" commands are not allowed. ⌟

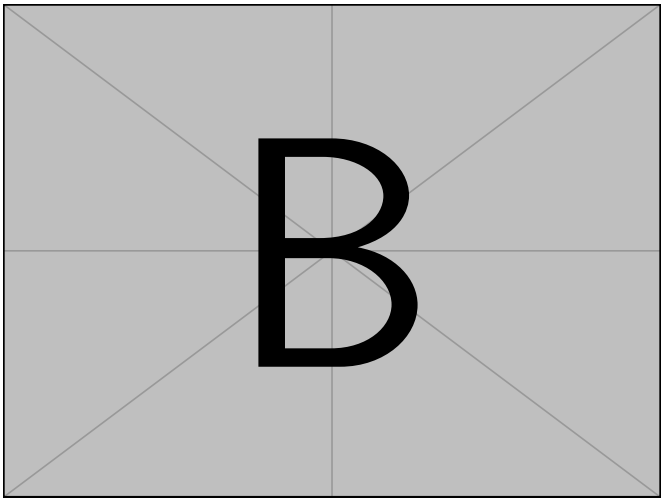

Figure A.5 / Example of a figure with a multi-paragraph caption.

Notice the spacing between the paragraphs. It was customized using the `parskip` key in `\captionsetup` provided by the `caption` → CTAN package.

To typeset verbatim text in the caption, use the `\fakeverb{...}` command instead of the usual `\verb|...|`, which is not allowed in captions.

A.1 Appendix Section

Note the numbering of various environments in the appendix.

Definition A.6 (Math in the Description — $\sin(\alpha) \approx \alpha$)**.** This is an example definition in an Appendix. Note the automatic switch to the alternative sans math font in the Definition description.

Remark A.7. The page header reflects that this is an appendix page. ⌟

References

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